

Bias and Missing Data:

The GradCafe data is voluntary, which introduces selection bias. Applicants who were accepted or had strong test scores (GPA, GRE) are more likely to share their results, while those with weaker outcomes or lower scores often remain silent. This explains why the average GRE Quantitative score in our dataset (177) is much higher than the national average (~157). It also explains the high average GPA (3.98). Missing data is common: many entries lack GRE scores, GPA, or comments. My scraper stored NULL for missing fields, preserving the integrity of the analysis.

Specific Findings and Limitations:

My dataset contained no entries for Johns Hopkins University in the program or `llm_generated_university` fields. This may be because: (1) my scraper did not capture those entries, (2) the LLM standardization failed to map the raw text to 'Johns Hopkins', or (3) no JHU CS Master's applicants reported to GradCafe during our scrape. Similarly, queries 7 and 8 returned zero results. This does not prove an absence of data but rather reveals limitations of our data cleaning process. The LLM-generated standardization of university and program names likely failed to map raw user entries to the specific canonical strings we queried for. This highlights a core limitation of using automated parsing on free-text, self-reported data—nuance and formatting variations can easily cause false negatives, even when relevant data exists in the source.